

Course Project Report: Environmental Sound Classification (ESC) via Audio Spectrogram Transformer

- **Course:** Building the Future of Voice & Audio
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1. Executive Summary

This report presents the design, implementation, and evaluation of an end-to-end Environmental Sound Classification (ESC) pipeline. Traditional models for classifying non-speech and non-music audio signals rely heavily on 2D Convolutional Neural Networks (CNNs). While effective, CNNs are constrained by local receptive fields, making it difficult to capture global temporal-spectral dependencies without extremely deep architectures.

To overcome these limitations, this project implements the Audio Spectrogram Transformer (AST)—a purely attention-based model—applied to the benchmark ESC-50 dataset. By adopting a memory-efficient linear probing strategy, the massive pre-trained AST encoder weights (86.2 million parameters) are frozen, and a custom 50-class classification head is trained. The pipeline includes programmatically downloading the dataset, auditing metadata, standardized digital signal processing (resampling, mono-mixing, log-Mel spectrogram conversion), training with state-of-the-art telemetry (TensorBoard & CSV logging), and a final validation on unseen test data.

On the independent, held-out test fold (Fold 5), the model achieved a Test Accuracy of 84.00% and a Macro F1-Score of 0.8247. A 50x50 confusion matrix was generated to analyze acoustic class overlaps, revealing mechanical and transient click confusions. Finally, a production-grade Command-Line Interface (CLI) tool was built to run real-world audio files through the identical DSP-model pipeline, verifying the system's deployment capability.

2. Problem Statement & Motivation

- The objective of this project is to categorize 50 distinct classes of environmental sounds (such as animal calls, natural sounds, water, domestic sounds, and urban noises) using an attention-based neural network. Unlike speech or music, environmental sounds:
 1. **Lack rigid sequential structure:** A dog bark or gunshot is a sudden transient; rain is a continuous stationary noise.
 2. **Contain highly diverse acoustic patterns:** Signals range from harmonic stacks (e.g., crying baby) to chaotic noise (e.g., wind).

- Traditional CNNs process spectrograms as standard images. However, convolutional kernels operate locally and struggle to capture long-range relationships between distant frequency bands or time intervals without stacking many layers.

3. Digital Signal Processing (DSP) Preprocessing

To match the inductive bias of the pre-trained AST model, raw audio files must be standardized.

- DSP Preprocessing Workflow:**
 - 1. Raw WAV (44.1 kHz, Stereo):** Loaded using torchaudio or soundfile.
 - 2. Mono Conversion:** If stereo, channels are averaged:

$$x_{\text{mono}}(t) = \frac{x_{\text{left}}(t) + x_{\text{right}}(t)}{2}$$

3. Resampling: Downsamples files from 44.1 kHz to 16 kHz to align with the AST backbone. 4. Length Standardization: Clips are forced to exactly 5.0 seconds (80,000 samples at 16 kHz) via zero-padding (if shorter) or truncation (if longer). 5. Log-Mel Transform:

- Window Size (N_{fft}):**

$$N_{\text{fft}}$$

Window Size (N_{fft}): 400 samples (25 ms duration) to balance frequency and time resolution.

- Hop Length (H):** 160 samples (10 ms stride) to capture transient audio features.
- Mel Bands (M):** 128 channels covering the human auditory range.
- DB Scale:** Linear power spectrograms are converted to decibel scale:

$$S_{\text{dB}} = 10 \log_{10} \left(\frac{S_{\text{mel}}}{\text{ref}} \right)$$

6. **Acoustic Normalization:** The resulting 128 × 501 spectrogram is normalized using AudioSet population statistics to accelerate training stability:

$$\hat{s}_{i,j} = \frac{s_{i,j} - \mu}{\sigma} \quad \text{where} \quad \mu = -4.2677, \sigma = 4.5690$$

4. Model Architecture & Linear Probing

The pre-trained AST model (MIT/ast-finetuned-audioset-10-10-0.4593) is adapted for the 50-class classification task.

- **Spectrogram Division:** The normalized spectrogram (1024×128) is divided into 16×16 overlapping patches, producing a sequence of 1212 tokens.
- **Token Projection:** Tokens are projected into a 768-dimensional space (AST hidden size).
- **CLS Token:** A trainable [CLS] classification token is prepended to the sequence.
- **Positional Embeddings:** 2D positional embeddings are added to preserve time-frequency coordinates.
- **Transformer Backbone:** The sequence is passed through a 12-layer Transformer Encoder.
- **Linear Classifier:** The output [CLS] token representation is sent to a custom linear classification head ($768 \rightarrow 50$).

Linear Probing Rationale:

Training an 86-million parameter transformer from scratch on a small dataset (2,000 samples) causes severe overfitting. **To prevent this, we freeze the transformer encoder backbone:**

$$\forall w \in W_{\text{backbone}}, \quad \frac{\partial \mathcal{L}}{\partial w} = 0$$

Only the weights of the newly attached classification head are updated during backpropagation (39,986 trainable parameters):

$$Y = W_{\text{classifier}} \cdot X + b$$

This reduces the computational footprint, enabling fast training and regularizing the model.

5. Training Infrastructure & Optimization

The pipeline utilizes a 5-fold cross-validation scheme to prevent data leakage. In this implementation:

- Training Folds: Folds 1, 2, and 3 (1,200 samples)
- Validation Fold: Fold 4 (400 samples)
- Test Fold: Fold 5 (400 samples)

Optimization Settings:

- **Loss Function:** Cross-Entropy Loss:

$$\mathcal{L} = - \sum_{c=1}^{50} y_c \log \left(\hat{y}_c \right)$$

- **Optimizer:** AdamW with a learning rate of 10^{-4} and weight decay of 10^{-4} .
- **LR Scheduler:** Cosine Annealing scheduler to decay the learning rate to 0 by the end of training.
- **Early Stopping:** Monitored on validation loss with a patience of 3 epochs.
- **Batch Size:** 16.

MLOps Telemetry:

We implemented dual-logging telemetry:

1. **CSV Logging:** Epoch-level loss and accuracy metrics are stored in `outputs/logs/training_history.csv` for flat-file plotting.
2. **TensorBoard Integration:** Step-level metrics, learning rates, and gradient statistics are logged dynamically.

6. Experimental Results & Test Evaluation

After training, the best checkpoint was loaded and evaluated on the completely unseen Fold 5 (400 samples).

Calculated Performance Metrics:

- **Test Loss:** 1.5185
- **Test Accuracy:** 84.00%
- **Macro Precision:** 0.8401
- **Macro Recall:** 0.8400
- **Macro F1-Score:** 0.8247
- **Weighted F1-Score:** 0.8247

Note: The Macro and Weighted scores are identical because the ESC-50 dataset is perfectly balanced, allocating exactly 8 samples per class per fold (400 total samples / 50 classes = 8).

7. Error Diagnosis & Acoustic Confusions

A 50x50 confusion matrix was generated to analyze class-level errors, saved as outputs/confusion_matrix.png . **The top 5 acoustic confusions are analyzed below:**

True Class	Predicted Class	Count	Acoustic Reason
wind	train	4	Continuous broadband pink noise profiles overlap with far-away diesel trains.
mouse_click	keyboard_typing	4	Short-duration, high-frequency transients (clicks) with identical decay times.
helicopter	engine	3	Rhythmic mechanical vibrations that mimic general combustion engine spectra.
washing_machine	vacuum_cleaner	3	Steady mechanical hums and continuous internal rotational noise.
airplane	helicopter	3	Doppler-shifting broadband noise and low-frequency resonance.

8. Command-Line Interface (CLI) Demonstration

To test the model in production, a command-line interface was developed to process individual audio files and output top predictions.

Command:

```
python src/predict.py dataset/ESC-50-master/audio/1-100032-A-0.wav
```

Outputs:

```
Preprocessing audio: dataset/ESC-50-master/audio/1-100032-A-0.wav
Loading model checkpoint: outputs/checkpoints/best_model.pt
```

```
=====
          ESC SOUND CLASSIFICATION RESULTS
=====
File: dataset/ESC-50-master/audio/1-100032-A-0.wav
-----
PREDICTED CLASS: DOG (Confidence: 17.21%)
-----
Top 5 Predictions:
1. dog                : 17.21%
2. crow               : 9.24%
3. sneezing           : 6.94%
4. cow                : 5.64%
5. thunderstorm       : 2.45%
=====
_____
```

9. Conclusion & Future Directions

This project successfully demonstrates the applicability of the Audio Spectrogram Transformer to environmental sound classification. Under a linear probing constraint (freezing 99.9% of weights), the model achieved a strong baseline of 84.00% test accuracy on unseen data.

To further improve performance:

- 1. Full Fine-Tuning:** Slowly unfreeze the transformer backbone layers using discriminative learning rates to adjust attention patterns.
- 2. Acoustic Augmentations:** Apply SpecAugment (masking random blocks of time and frequency channels) to prevent overfitting during full training.
- 3. Multi-Fold Cross-Validation:** Train 5 separate model checkpoints (one for each validation fold) and ensemble their predictions during inference.

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